LESSON 7.3

Understand Absolute Value of Rational Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the absolute value of a rational number — an integer, fraction, or decimal — as its distance from zero.

WHY THIS MATTERS Claire is performing a science experiment with two liquids. The temperatures of the two liquids in Celsius are shown. She needs to use the beaker with the temperature that is closer to 0°C. Which beaker should Claire use?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	... -3, -2, -1, 0, 1, 2, 3 ... shown on a number line with 0 in the center
Positive *Spanish: Positivo*	A number bigger than zero, to the right of zero on a number line.	+5 means 5 above zero, like 5 floors above ground level
Negative *Spanish: Negativo*	A number smaller than zero, to the left of zero on a number line.	-3 means 3 below zero, like 3 floors underground
Absolute value *Spanish: Valor absoluto*	How far a number is from zero. It is never negative.	$\|-3\| = 3$ and $\|3\| = 3$
Opposite *Spanish: Opuesto*	Two numbers the same distance from zero, on opposite sides, like 3 and -3.	−3 and 3 are opposites
Rational number *Spanish: Número racional*	A number that can be written as a fraction of two integers, with a bottom number that is not zero.	−3/4, 0.5, 7, −50

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find $|-13|$ and $|6|$. Which number is farther from 0?

1. −13 is 13 units left of 0, so $|-13| = 13$.
2. 6 is 6 units right of 0, so $|6| = 6$.
3. Compare the distances: $13 > 6$.

Answer: −13 is farther from 0

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's find |3|. Where is 3 on the number line? It is 3 steps to the right of zero.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The absolute value of _____ is _____ because it sits _____ units from zero.
- _____ and _____ share the same absolute value because _____ .
- An absolute value is never negative because _____ .
- The number closer to zero is _____ because its distance is _____ .

Word bank absolute value | distance | magnitude | zero | number line | opposite | units | closer | farther | negative | positive | sign

WATCH OUT!

A common mistake

A common mistake in Integers and Absolute Value is judging size by the digits alone instead of position on the number line

LESSON 7.3

Understand Absolute Value of Rational Numbers

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each integer to its absolute value.

_____ -5	(A) 5
_____ 3	(B) 3
_____ -8	(C) 8
_____ 0	(D) 0
_____ -1	(E) 1
_____ 7	(F) 7

HINT Each integer matches its distance from zero. Negative signs disappear; positives stay the same.

2 **Mark and label each value on the number line.**

Plot these integers on the number line: -6, -2, 0, 4, 7



HINT Negatives go LEFT of 0 and positives go RIGHT. Zero sits in the middle.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the absolute value of -9?

(A) -9
(B) 0
(C) 1
(D) 9

HINT Absolute value asks: how FAR is -9 from zero? It's about distance, not direction.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

(A) -5
(B) -3
(C) 2
(D) 7

HINT 'Greatest absolute value' means farthest from zero — the sign doesn't matter.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is |0|?

(A) 1
(B) -1
(C) 0
(D) There is no answer

HINT Absolute value = distance from zero. Here the number inside the bars IS zero.

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

What is |−2.5|?

(A) 0

(B) 2.5

(C) −2.5

(D) 25

HINT Where does −2.5 sit on the number line?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does the number line help you see the difference between an integer and its absolute value?

SENTENCE STARTER The integer ___ is located ___ units from zero, so its absolute value is ___. Even though ___ is negative, its distance from zero is still ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.3

Understand Absolute Value of Rational Numbers

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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WHY THIS MATTERS Claire is performing a science experiment with two liquids. The temperatures of the two liquids in Celsius are shown. She needs to use the beaker with the temperature that is closer to 0°C. Which beaker should Claire use?

PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Plot these integers and circle pairs that have the same absolute value: -6, 6, -2, 2, -4, 4



SHOW YOUR THINKING

2 **Complete the table. Show how you found each value.**

Complete the table with each integer's absolute value and its opposite.

Integer	Absolute Value	Opposite
-7		
4		
-12		
0		
9		

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Sort these integers from the GREATEST absolute value to the LEAST absolute value.

Groups:

_____ -2

_____ 8

_____ -5

_____ 1

_____ -10

_____ 3

4 **Circle the letter of the best answer. Show how you know.**

Which statement about absolute value is TRUE?

(A) |-4| = |4| because both are 4 units from zero

(B) |-4| = -4 because the number is negative

(C) |-4| = 0 because negatives cancel out

(D) |4| > |-4| because positive is always bigger

SHOW YOUR WORK

PART 2 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

Which number has the LEAST absolute value: −1.5, 0.75, −1/4, or 2?

(A) 0.75
(B) −1/4
(C) −1.5
(D) 2

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Confusing Opposite with Absolute Value

1. **Problem** Complete the table for the integer 9: find its absolute value and its opposite.
2. **Student's thinking** Absolute value and opposite both just make a number positive, so they should give the same answer as each other.
3. **Student's answer** Absolute value of 9 = 9, and opposite of 9 = 9.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does the number line help you see the difference between an integer and its absolute value?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.3

Understand Absolute Value of Rational Numbers

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Decide whether each pair balances. Check Yes or No, then explain.**

Use absolute value to decide which side is greater, or if they are equal.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the Absolute Value Error

1. **Problem** Find |-6|
2. **Student thinking** The number is -6, and absolute value makes it the opposite
3. **Student answer** |-6| = 6, and since we take the opposite, the answer is -6

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

A submarine is at -150 meters and a bird is flying at +150 meters. A student says the submarine is closer to sea level because -150 is less than 150. Is the student correct? Explain using absolute value.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How does the number line help you see the difference between an integer and its absolute value?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help